

FLAW SIGNALS

ON THE LISSAJOUS

What a real flaw looks like on a bobbin differential, and how it should behave.

KEY IDEA

Angle tells you depth. Voltage tells you size. A real flaw is clean, closed, repeatable, and shows on more than one channel.

1 THE SHAPE

- Differential: a figure-8, with one loop for each coil as it passes the flaw.
- A clean flaw is smooth, closed, and fairly even. It starts and ends at null.
- Absolute: a single loop. Better for gradual change like wear or thinning.
- Differential is better for sudden change like pits and cracks.

2 READING THE ANGLE

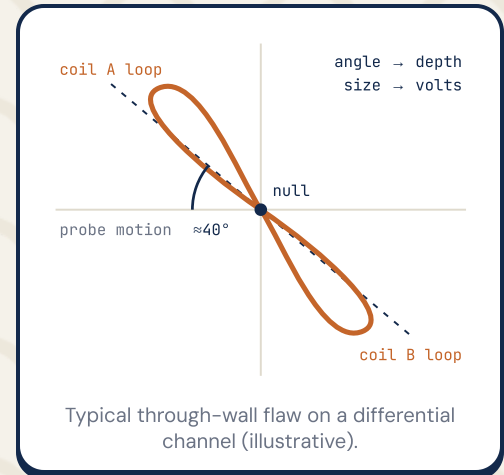
- A through-wall hole sits on your reference line (about 40° on most setups).
- OD flaws rotate past that line as they get shallower.
- ID flaws sit on the other side of the line.
- Decide ID or OD before you read depth off the curve.

3 HOW A REAL FLAW BEHAVES

- Shows on more than one frequency, at the same axial spot.
- Traces the same way as your cal flaws.
- Repeats on a rescan. Noise doesn't.
- Doesn't lie flat along the probe-motion line. Flat is usually wobble or lift-off.

4 WHICH FREQUENCY SEES WHAT

- High frequency (CH1) is most sensitive at the ID surface.
- Lower frequencies reach deeper into the wall and help with OD flaws.
- Very low frequency is for supports, deposits, and landmarks, not sizing.



RED FLAGS

- ✗ Messy or open shape that doesn't return to null
- ✗ Shows on one frequency only
- ✗ Mostly horizontal
- ✗ Changes from scan to scan
- ✗ Lobes mirrored compared with the cal flaws

QUICK TIPS

Compare every call with the same flaw type on the cal standard.

Size on the channel your procedure names.

Not sure? Flag it. Don't guess.